

Tasca Avaluable 7.4

1. Carrega la informació dels arxius Transactions.json i Accounts.json al mysql workbench en dues taules. Explica tot el procés amb captures de pantalla i copia el codi.

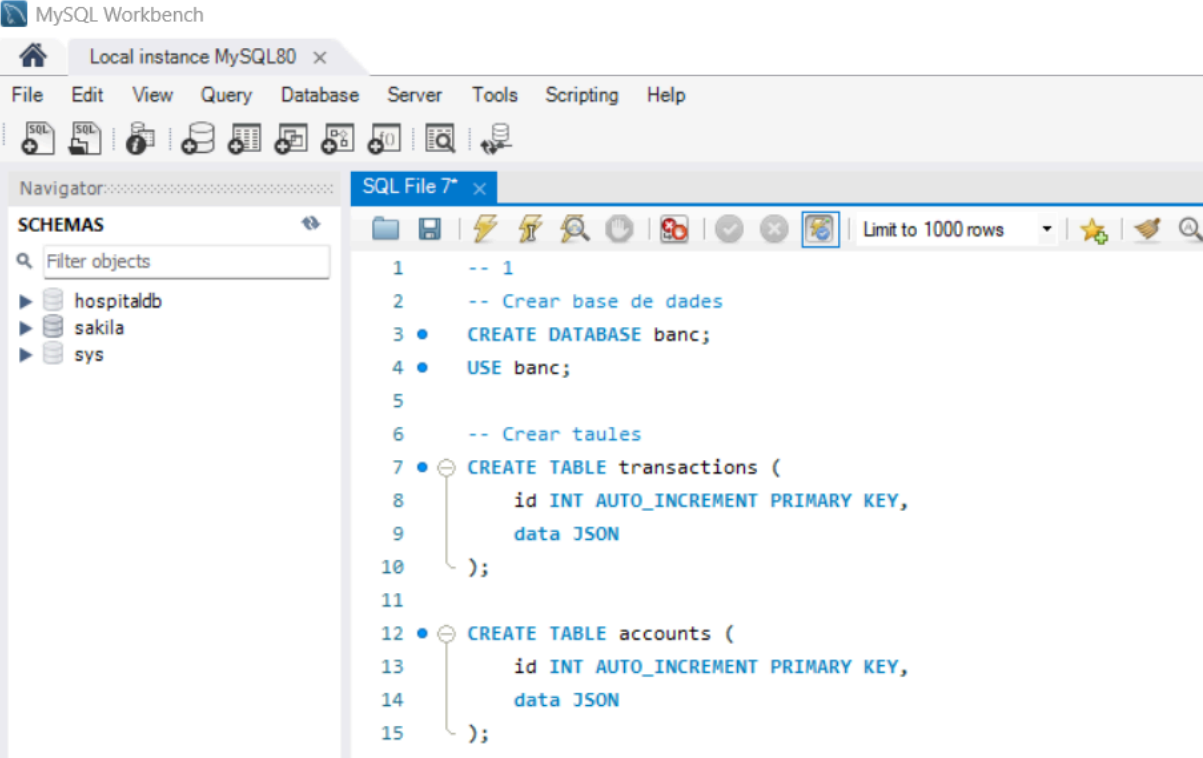
```
-- 1
-- Crear base de dades
CREATE DATABASE banc;
USE banc;

-- Crear taules
CREATE TABLE transactions (
  id INT AUTO_INCREMENT PRIMARY KEY,
  data JSON
);

CREATE TABLE accounts (
  id INT AUTO_INCREMENT PRIMARY KEY,
  data JSON
);

-- Importar les dades
INSERT INTO transactions (data)
VALUES (
  CONVERT(
    LOAD_FILE('C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/Transactions.json')
    USING utf8mb4
  )
);

INSERT INTO accounts (data)
VALUES (
  CONVERT(
    LOAD_FILE('C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/Accounts.json')
    USING utf8mb4
  )
);
```



MySQL Workbench

Local instance MySQL80 x

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Navigator

SCHEMAS

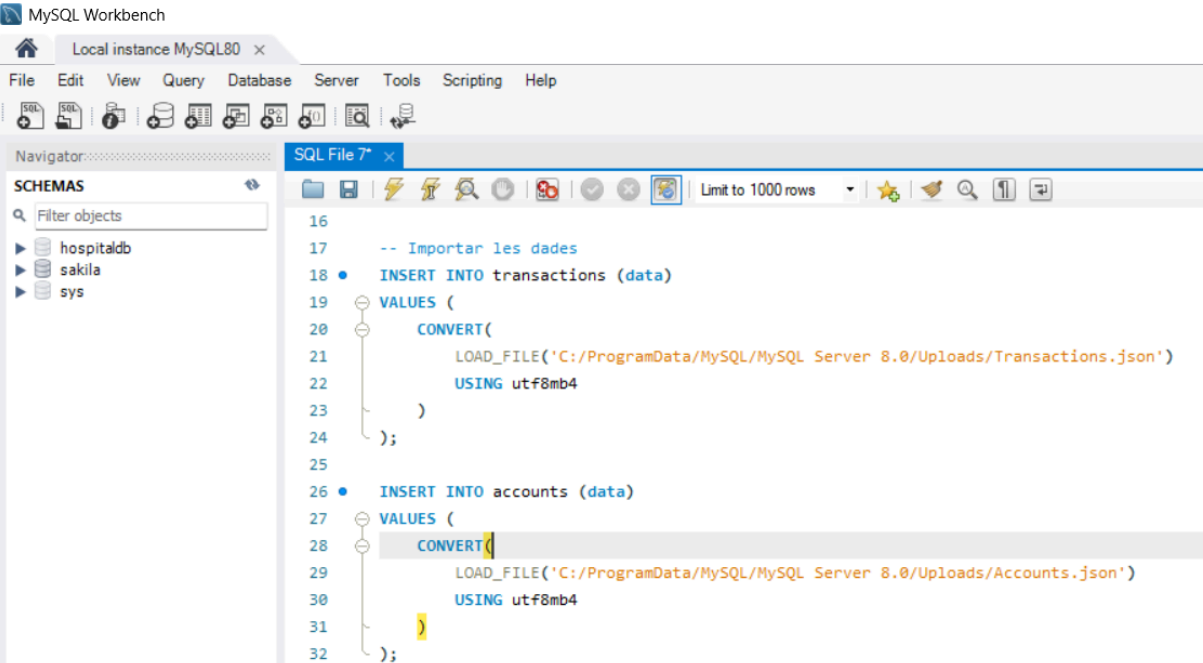
Filter objects

- hospitaldb
- sakila
- sys

SQL File 7* x

Limit to 1000 rows

```
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2  -- Crear base de dades
3  • CREATE DATABASE banc;
4  • USE banc;
5
6  -- Crear taules
7  • CREATE TABLE transactions (
8      id INT AUTO_INCREMENT PRIMARY KEY,
9      data JSON
10 );
11
12 • CREATE TABLE accounts (
13     id INT AUTO_INCREMENT PRIMARY KEY,
14     data JSON
15 );
16
```



MySQL Workbench

Local instance MySQL80 x

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SQL File 7* x

Limit to 1000 rows

```
16
17 -- Importar les dades
18 • INSERT INTO transactions (data)
19 VALUES (
20     CONVERT(
21         LOAD_FILE('C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/Transactions.json')
22         USING utf8mb4
23     )
24 );
25
26 • INSERT INTO accounts (data)
27 VALUES (
28     CONVERT(
29         LOAD_FILE('C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/Accounts.json')
30         USING utf8mb4
31     )
32 );
```

1.1. Quina consulta utilitzaries per obtenir l'import (amount) de totes les transaccions de la taula transactions (Utilitza la funció JSON_UNQUOTE per a eliminar les cometes)?

```
SELECT
  JSON_UNQUOTE(JSON_EXTRACT(jt.txn, '$.amount')) AS amount
FROM transactions,
JSON_TABLE(
  data,
  '$.transactions[*]' COLUMNS (
    txn JSON PATH '$'
  )
) AS jt;
```

```
34      -- 1.1
35      SELECT
36          JSON_UNQUOTE(JSON_EXTRACT(jt.txn, '$.amount')) AS amount
37      FROM transactions,
38      JSON_TABLE(
39          data,
40          '$.transactions[*]' COLUMNS (
41              txn JSON PATH '$'
42          )
43      ) AS jt;
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	amount			
▶	100			
	500			
	75			
	1125			
	250			
	325			
	350			

Result 21 x

1.2. Com podries filtrar les transaccions que tenen una quantitat superior a 200 (utilitza JSON_EXTRACT)?

```
SELECT
  JSON_UNQUOTE(JSON_EXTRACT(jt.txn, '$.txn_id')) AS txn_id,
  JSON_OBJECT(
    'amount', CAST(JSON_EXTRACT(jt.txn, '$.amount') AS UNSIGNED),
    'txn_date', JSON_UNQUOTE(JSON_EXTRACT(jt.txn, '$.txn_date'))
  ) AS txn_data
FROM transactions,
JSON_TABLE(
  data,
  '$.transactions[*]' COLUMNS (
    txn JSON PATH '$'
  )
) AS jt
WHERE CAST(JSON_EXTRACT(jt.txn, '$.amount') AS DECIMAL(10,2)) > 200;
```

```
45      -- 1.2
46 •    SELECT
47          JSON_UNQUOTE(JSON_EXTRACT(jt.txn, '$.txn_id')) AS txn_id,
48          JSON_OBJECT(
49              'amount', CAST(JSON_EXTRACT(jt.txn, '$.amount') AS UNSIGNED),
50              'txn_date', JSON_UNQUOTE(JSON_EXTRACT(jt.txn, '$.txn_date'))
51          ) AS txn_data
52      FROM transactions,
53      JSON_TABLE(
54          data,
55          '$.transactions[*]' COLUMNS (
56              txn JSON PATH '$'
57          )
58      ) AS jt
59      WHERE CAST(JSON_EXTRACT(jt.txn, '$.amount') AS DECIMAL(10,2)) > 200;
60
```

result Grid	
Filter Rows:	Export: Wrap Cell Content:
txn_id	txn_data
2	{"amount": 500, "txn_date": "2000-01-15 00:0...
4	{"amount": 1125, "txn_date": "2001-03-12 00:...
5	{"amount": 250, "txn_date": "2001-03-12 00:0...
6	{"amount": 325, "txn_date": "2002-11-23 00:0...
7	{"amount": 350, "txn_date": "2002-12-15 00:0...
10	{"amount": 452, "txn_date": "2004-09-30 00:0...

1.3. Escriu una consulta que obtingui l'ID de la transacció (txn_id) i la data de la transacció (txn_date) per totes les transaccions.

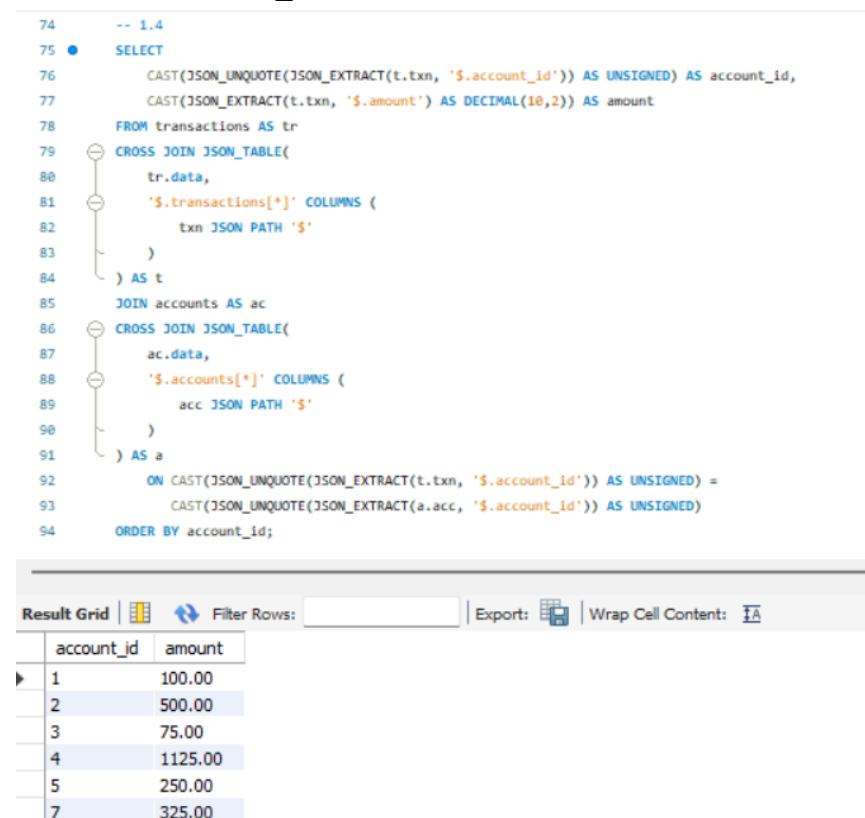
```
SELECT
  JSON_UNQUOTE(JSON_EXTRACT(jt.txn, '$.txn_id')) AS txn_id,
  JSON_UNQUOTE(JSON_EXTRACT(jt.txn, '$.txn_date')) AS txn_date
FROM transactions,
JSON_TABLE(
  data,
  '$.transactions[*]' COLUMNS (
    txn JSON PATH '$'
  )
) AS jt
ORDER BY CAST(JSON_UNQUOTE(JSON_EXTRACT(jt.txn, '$.txn_id')) AS UNSIGNED);
```

```
61      -- 1.3
62 •    SELECT
63        JSON_UNQUOTE(JSON_EXTRACT(jt.txn, '$.txn_id')) AS txn_id,
64        JSON_UNQUOTE(JSON_EXTRACT(jt.txn, '$.txn_date')) AS txn_date
65      FROM transactions,
66      JSON_TABLE(
67        data,
68        '$.transactions[*]' COLUMNS (
69          txn JSON PATH '$'
70        )
71      ) AS jt
72      ORDER BY CAST(JSON_UNQUOTE(JSON_EXTRACT(jt.txn, '$.txn_id')) AS UNSIGNED);
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
txn_id	txn_date			
1	2000-01-15 00:00:00			
2	2000-01-15 00:00:00			
3	2004-06-30 00:00:00			
4	2001-03-12 00:00:00			
5	2001-03-12 00:00:00			
6	2002-11-23 00:00:00			
7	2002-12-15 00:00:00			

1.4. Escriu una consulta que recuperi l'ID del compte (account_id) i l'import (amount) de les transaccions, unint les taules transactions i accounts per account_id.

```
SELECT
  CAST(JSON_UNQUOTE(JSON_EXTRACT(t.txn, '$.account_id')) AS UNSIGNED) AS
account_id,
  CAST(JSON_EXTRACT(t.txn, '$.amount') AS DECIMAL(10,2)) AS amount
FROM transactions AS tr
CROSS JOIN JSON_TABLE(
  tr.data,
  '$.transactions[*]' COLUMNS (
    txn JSON PATH '$'
  )
) AS t
JOIN accounts AS ac
CROSS JOIN JSON_TABLE(
  ac.data,
  '$.accounts[*]' COLUMNS (
    acc JSON PATH '$'
  )
) AS a
ON CAST(JSON_UNQUOTE(JSON_EXTRACT(t.txn, '$.account_id')) AS UNSIGNED) =
  CAST(JSON_UNQUOTE(JSON_EXTRACT(a.acc, '$.account_id')) AS UNSIGNED)
ORDER BY account_id;
```



```
74 -- 1.4
75 SELECT
76   CAST(JSON_UNQUOTE(JSON_EXTRACT(t.txn, '$.account_id')) AS UNSIGNED) AS account_id,
77   CAST(JSON_EXTRACT(t.txn, '$.amount') AS DECIMAL(10,2)) AS amount
78 FROM transactions AS tr
79 CROSS JOIN JSON_TABLE(
80   tr.data,
81   '$.transactions[*]' COLUMNS (
82     txn JSON PATH '$'
83   )
84 ) AS t
85 JOIN accounts AS ac
86 CROSS JOIN JSON_TABLE(
87   ac.data,
88   '$.accounts[*]' COLUMNS (
89     acc JSON PATH '$'
90   )
91 ) AS a
92 ON CAST(JSON_UNQUOTE(JSON_EXTRACT(t.txn, '$.account_id')) AS UNSIGNED) =
93   CAST(JSON_UNQUOTE(JSON_EXTRACT(a.acc, '$.account_id')) AS UNSIGNED)
94 ORDER BY account_id;
```

account_id	amount
1	100.00
2	500.00
3	75.00
4	1125.00
5	250.00
7	325.00

1.5. Com podries calcular l'import total de les transaccions per cada compte (account_id)?

```
SELECT
  CAST(JSON_UNQUOTE(JSON_EXTRACT(jt.txn, '$.account_id')) AS UNSIGNED) AS
  account_id,
  SUM(CAST(JSON_EXTRACT(jt.txn, '$.amount') AS DECIMAL(10,2))) AS total_amount
FROM transactions AS tr
CROSS JOIN JSON_TABLE(
  tr.data,
  '$.transactions[*]' COLUMNS (
    txn JSON PATH '$'
  )
) AS jt
GROUP BY CAST(JSON_UNQUOTE(JSON_EXTRACT(jt.txn, '$.account_id')) AS
UNSIGNED)
ORDER BY account_id;
```

```
96  -- 1.5
97  •  SELECT
98      CAST(JSON_UNQUOTE(JSON_EXTRACT(jt.txn, '$.account_id')) AS UNSIGNED) AS account_id,
99      SUM(CAST(JSON_EXTRACT(jt.txn, '$.amount') AS DECIMAL(10,2))) AS total_amount
100  FROM transactions AS tr
101  CROSS JOIN JSON_TABLE(
102      tr.data,
103      '$.transactions[*]' COLUMNS (
104          txn JSON PATH '$'
105      )
106  ) AS jt
107  GROUP BY CAST(JSON_UNQUOTE(JSON_EXTRACT(jt.txn, '$.account_id')) AS UNSIGNED)
108  ORDER BY account_id;
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
account_id	total_amount			
1	100.00			
2	500.00			
3	75.00			
4	1125.00			
5	250.00			
7	325.00			
8	350.00			

Enllaç Github: https://github.com/yanalexanderadames-gif/Llenguatge_De_Marques